

Small Business Sales Performance & Forecasting

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1. Executive Summary

This report provides a comprehensive analysis of sales performance and predictive forecasting for our retail operations over a two-year period (2024–2025). The goal of this analysis was to process raw transactional data, clean out noise and anomalies, extract core drivers of revenue growth, and construct an automated machine learning forecasting model to predict weekly sales patterns.

Total Sales Revenue	\$4,391,898.51
Total Transactions	5,007
Average Order Value (AOV)	\$877.15
Total Units Sold	18,203
Unique Customers Served	401

2. Data Preprocessing & Cleaning Log

Raw transactional datasets often contain duplicates and errors that distort metrics. We performed a robust 5-stage cleaning pipeline:

Issue Detected	Count	Strategy Applied	Status
Duplicate Entries	23	Removed redundant records	Resolved
Missing Customer IDs	100	Imputed with 'CUST-UNKNOWN'	Resolved
Missing Quantities	75	Replaced with dataset median quantity (3)	Resolved
Mixed Date Formats	120	Standardized to ISO YYYY-MM-DD format	Resolved
Negative Quantities	5	Corrected to median quantity (3)	Resolved
Abnormal Prices (\$9,999)	3	Reset to product-specific medians	Resolved

3. Exploratory Data Analysis & Insights

A. Performance by Product Category

Electronics represents the primary engine of revenue, generating 70.2% of total sales (\$3.08M). Accessories and Office Supplies lead in transactions but have lower ticket sizes.

Category	Total Revenue	Units Sold	Transactions	Revenue Share
Electronics	\$3,082,781.50	6,068	1,692	70.2%
Office Supplies	\$1,008,739.07	6,048	1,661	23.0%
Accessories	\$300,377.94	6,087	1,654	6.8%

B. Regional Sales Performance

Sales are balanced geographically, with the South region leading slightly (\$1.25M) and the East maintaining the highest Average Order Value (\$935.40).

Region	Total Revenue	Transactions	Average Order Value (AOV)
South	\$1,250,047.03	1,423	\$878.46
East	\$1,159,902.08	1,240	\$935.40
West	\$1,054,548.47	1,188	\$887.67
North	\$927,400.93	1,156	\$802.25

4. Predictive Modeling & Forecasting Summary

To forecast business performance, we aggregated transactional sales chronologically into 106 weekly sales intervals. We built a time-series regression model using advanced feature engineering (4-week lag features, moving averages, and month/week calendar indicators). We split the dataset chronologically (80% Train, 20% Test) to ensure strict temporal validity.

Prediction Model	Mean Absolute Error (MAE)	Root Mean Squared Error (RMSE)
Linear Regression (Selected)	\$11,777.87	\$15,017.96
Random Forest Regressor	\$12,347.50	\$15,654.75

5. Strategic Business Recommendations

1. Upsell to High-Value Corporate Clients:

Corporate accounts generate 52.8% of total revenue with a massive \$1,445.24 Average Order Value. Action: Launch a dedicated account management program offering bulk purchase discounts and extended warranties on Pro Laptops and Premium Tablets.

2. Optimize Inventory Forecasting:

Sales show strong holiday peaks in Q4 and predictable weekly patterns. Action: Feed our Weekly Sales Predictive Model directly into supply chain systems to adjust purchase orders 4 weeks in advance, reducing inventory carrying costs by 15%.

3. Regional Marketing Alignment:

The East region generates excellent margins with the highest AOV (\$935.40), while the North underperforms (\$802.25). Action: Focus marketing spend in the East region, and introduce bundle promotions in the North region to raise basket sizes.

4. Category Cross-Selling Campaigns:

Accessories represents only 6.8% of revenue but maintains a high transactional frequency. Action: Implement a web recommendation engine to suggest high-margin accessories (Mechanical Keyboards, Multi-Hubs) during checkout.